

Assignment 1

Foundations of Data Analytics and Data Science

Section A: Data Fundamentals

1. A Database is a smart digital filing cabinet where organized collections of data are stored electronically, managed and can be retrieved for specific company or business decision-making purposes.

They are essential in the modern world because they handle lots of information, allowing fast and accurate retrieval where many people can use the same data at once, keeping it consistent and supporting better business decisions.

Examples:

1. A school system (learner records, grades, attendance and academic history).
2. E-commerce/retail (orders, stock levels, customer information).

2. A Data Warehouse is a large central storage system that pulls data from different sources to analyze and spot trends over time. It differs from a regular database as it stores combined historical data for Online Analytical Processing (OLAP), where updates are not primarily made in real-time.

Companies use Data Warehouses to combine different systems, spot trends that help improve

efficiency long-term, and run big reports without slowing down daily operations

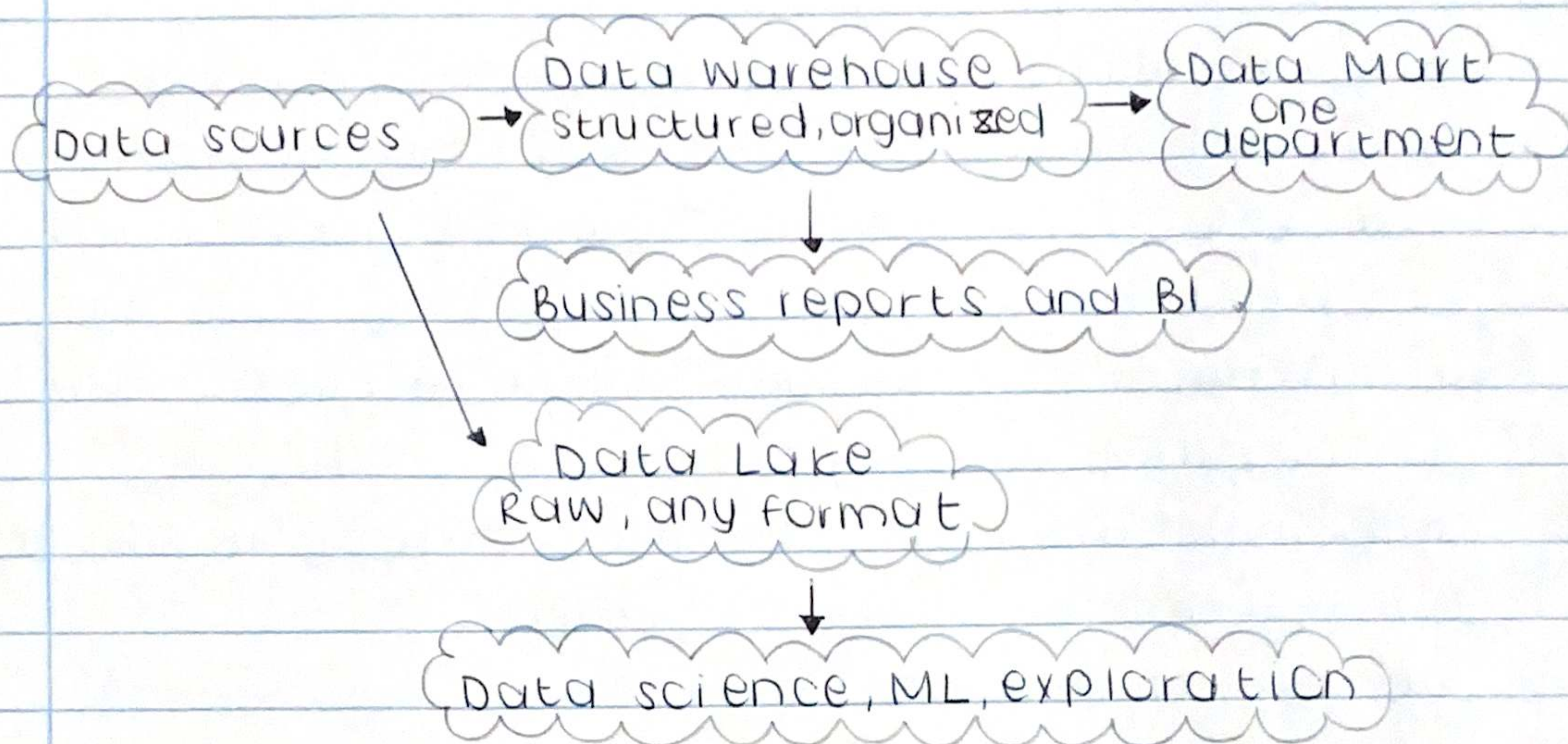
3. A Data Lake is a storage system where huge amounts of raw data are held. There are 3 different types thereof, namely, structured, semi-structured and unstructured. This data is not organized as it is stored exactly as it comes in.

The difference between a Data Lake and a Data Warehouse is that warehouses store cleaned, organized data for specific purposes, while Lakes store data exactly as it comes and can be cleaned or organized later when actually needing to use it.

Businesses prefer Lakes over warehouses as they are less expensive and flexible for storing huge raw volumes of data.

4. A Data Mart is a smaller version of a data warehouse that holds data relevant to just one department or team instead of the entire company as a whole. It relates to a Data Warehouse as it is often built from it and focuses on storing only relevant data for a specific department or team.

It is used by specific departments or teams that require quick, focused access to their own data instead of searching through the entire company data.



5. ① Structured data: Neatly organized into rows and columns.

Example: Excel sheet of customer names and contact numbers.

② Semi-structured: A bit organized but not as rigid as a table where tags or labels are commonly used.

Example: Email, has fields like sender/subject, but the body is free text.

③ Unstructured: No fixed format.

Example: Videos, photos, voice recordings

structured
e.g. spread
sheet

semi-structured
e.g. an email

subject: tagged
sender: tagged
body: free
text, no fixed form

unstructured
e.g. photo,
audio

photo

audio

6. Data Modeling is like a blueprint for data where the process of planning and designing how the data will be organized, stored and connected takes place before being built into a system. It's important as it ensures data is structured logically, errors are reduced and duplicates are avoided making it easier for systems to work together.

The 2 common types are:

① Conceptual model: Broad view of what data is required and how it relates to the other (less detailed).

② Relational/Logical or Physical model: Detailed, defines actual tables, fields, and how they connect in the database.

7. Data Mining is when businesses extract and discover new or pre-existing patterns, trends and useful information in large datasets. It helps them better understand customer behaviour, foretell future trends and make data-driven decisions.

Examples:

1. Banks detect unusual transactions that may signal fraud.

2. Retailers analyze purchase history to recommend products (Best-seller).

8. A Database Management System is a software tool that sits between me and the actual data stored and it allows one to create, manage

and interact with a database.

• Four popular DBMS's are:

1. MySQL: Free, open-source system that's good for creating websites and small-to-medium apps.
2. PostgreSQL: Very reliable, open-source and handles complex data, great for businesses that need advanced features.
3. Oracle: Paid enterprise system used for and by big companies that need high security and performance (banks)
4. MongoDB: NoSQL system storing flexible, unstructured-like data instead of strict tables, good for changing data needs, like social media platforms.

9. Extract, Transform and Load is the process of moving data from one place to another for analyzing.

- Extract: pulls raw data from its original source (POS)
- Transform: cleans and reformats data to ensure consistency and usability (removes duplicates, fixes errors).
- Load: stores transformed data where it's supposed to be, e.g.: Data warehouse.

This process is important for it ensures work is accurate, consistent and ready to use eliminating messy data which leads to bad business decisions.



10. ① CSV: Easy to use simple text file with data separated by commas. Not recommended for complex or huge amounts of data, commonly used for small datasets.

② JSON: Flexible, easy for systems to read and exchange, stores data with labeled fields and is good for semi-structured data.

③ Parquet: Used when working with big datasets and stores data in a compressed and efficient way that's quick to analyze.

④ Avro: Created for storing and moving data fast between systems, used in real-time pipelines.

► References:

1. IBM. (2026) What is a Database? Available at: <https://www.ibm.com/think/topics/database> (Accessed: 8 September 2026)
2. AWS. (2026) What's the difference between a Data warehouse, Data Lake, and Data Mart? Available at: <https://aws.amazon.com/compare/the-difference-between-a-data-warehouse-data-lake-and-data-mart> (Accessed: 8 September 2026)
3. IBM (2026) Structured vs Unstructured Data: What's the Difference? Available at: <https://www.ibm.com/think/topics/structured-vs-unstructured-data> (Accessed:

8 September 2026).

4. IBM. (2026) What is Data Modeling? Available at: <https://www.ibm.com/think/topics/data-modeling> (Accessed: 8 September 2026).
5. IBM. (2026) What is Data Mining? Available at: <https://www.ibm.com/think/topics/data-mining> (Accessed: 8 September 2026).

Section B: Artificial Intelligence & Machine Learning

11. AI is technology that allows computers to perform tasks that normally require human thinking, recognizing speech, solving problems and making decisions that are said common tasks.

- Narrow AI is designed to do one specific task really well, like a specialist.

Example: Siri.

- General AI is AI that can think, learn and perform many different tasks like a human.

Example: Iron Man's Jarvis

12. Machine Learning is a type of AI where computers learn patterns from data by studying examples instead of being given exact instructions by humans.

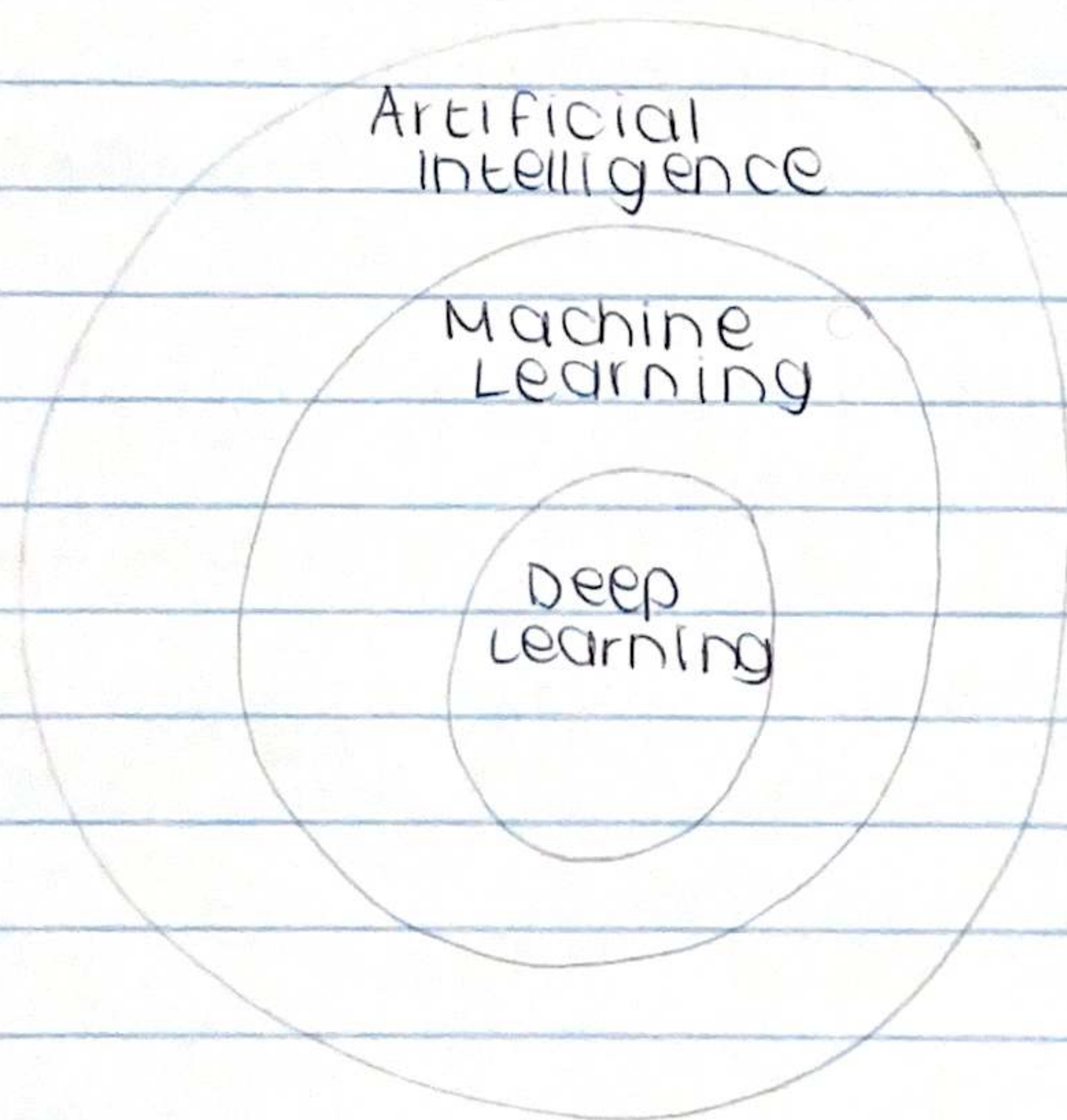
- Supervised learning: Learns from labelled data

- Unsupervised learning: Finds patterns in data with no labels.

- Reinforcement learning: Learns through trial and error by receiving rewards for good actions and penalties for bad ones.

13. Deep Learning is a more advanced type of Machine Learning that uses different layers to identify complex patterns and perform harder tasks like speech and image recognition. It relates to ML and AI in the sense that AI is the broad field, ML is a part of AI and DL is a part of ML.

Neural Networks are systems inspired by the human brain that uses interconnected nodes to process data and identify patterns, i.e. Face recognition on a phone.



14. NLP is how computers interpret and respond to human language (text or speech).

Examples:

- ① Autocorrect on my phone
- ② Voice assistants like Siri on my phone
- ③ Spam filters detecting junk emails.

15. A Large Language Model is a type of AI trained on huge amounts of text, helping it understand and generate human-like language. It differs from a standard ML model because it is often designed for a specific task, while an LLM can handle different language tasks such as answering questions, writing and summarizing, all in one model.

Examples:

1. ChatGPT: used for conversation, writing, answering questions and coding help.
2. Claude: used for chatting, research and analysis.

16. Generative AI doesn't just analyze existing data but creates new content. At a high level, it learns patterns from large amounts of existing content and uses those patterns to generate something new or similar in style.

Examples:

1. Midjourney - generates images from text descriptions
2. ElevenLabs - generates realistic AI audio/voice
3. ChatGPT/Claude - generates written text or answers.

17. A training dataset is data used to teach an AI model how to recognize patterns before they are used in the real world. The quality matters because AI only learns what is shown, so if the data is wrong, incomplete or biased, the AI can learn those same mistakes.

What could go wrong is if a hiring AI is trained mostly on data from one gender, it might unfairly favour that gender in future hiring decisions, even if that was never the true intent.

18. Computer vision is the manner in which machines look at and understand images or videos by analyzing pictures as pixels and patterns that AI learns to recognize.

Examples:

1. Face recognition to unlock a phone.
2. Waymo cars using computer vision to detect pedestrians, road signs, and other vehicles.

Section C: API's, tools & developer concepts

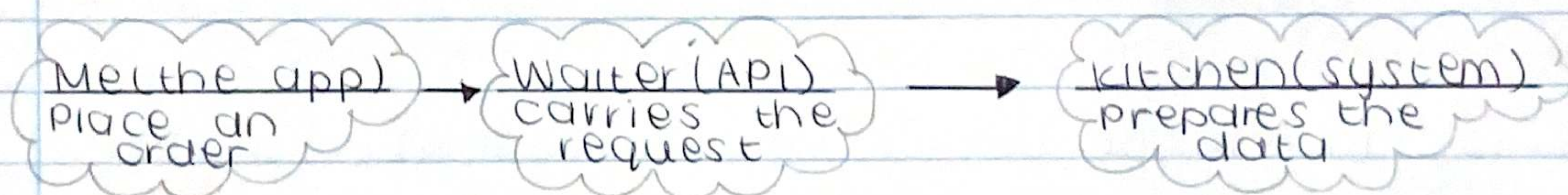
► References:

1. IBM. (2026) What is Artificial Intelligence (AI)? Available at: <https://www.ibm.com/think/topics/artificial-intelligence> (Accessed: 9 September 2026)
2. Google for Developers. (2026) Machine Learning. Available at: <https://www.developers.google.com/machine-learning> (Accessed: 9 September 2026).
3. IBM. (2026) What are LLM? Available at: <https://www.ibm.com/think/topics/large-language-models> (Accessed: 9 September 2026).

Section C: APIs, Tools and Developer concepts

19. An Application Programming interface is a set of rules that allows different applications "talk" to each other and exchange information, kind of like a waiter at a restaurant, I can't physically enter the restaurant kitchen, but the waiter can take my order and bring back what I've asked for.

APIs matter in modern software and data systems as they allow databases and apps to connect and share data without needing to know how the other system works internally.



20. An API key is a password that identifies me and grants me permission to use a specific API. Its purpose is to allow the provider to track and control my access. It should be stored and protected by keeping it secret and storing it in secure settings or environment variables.

It should never be posted publicly in code because if it is exposed publicly, someone else could use it, potentially run up charges or access data they shouldn't, keeping it secure is of utmost importance.

21. A REST API is a set of principles and rules that can be applied using any programming language. REST stands for Representational State Transfer, and the main HTTP methods used are:

- GET: This method retrieves data
- POST: This method creates new data
- PUT: This method updates existing data
- DELETE: This method removes data.



22. JavaScript Object Notation (JSON) is a text format used to store and/or share data using labelled fields. It is lightweight and easy for both humans and machines to read and understand, which is why it is commonly used when working with APIs and exchanging data.

Example:

```
{  
  "name": "Sinothando"  
  "role": "Customer Experience Specialist"  
  "city": "Pretoria"
```

23. Prompt Engineering is the skill of writing clear and specific instructions to receive the best response from an AI tool. If your prompt is clear

and specific, the AI is more likely to provide a useful and relevant result. It is an important skill when working with AI tools like chatgpt and claude because the quality of the response depends largely on the instructions or prompt that is given.

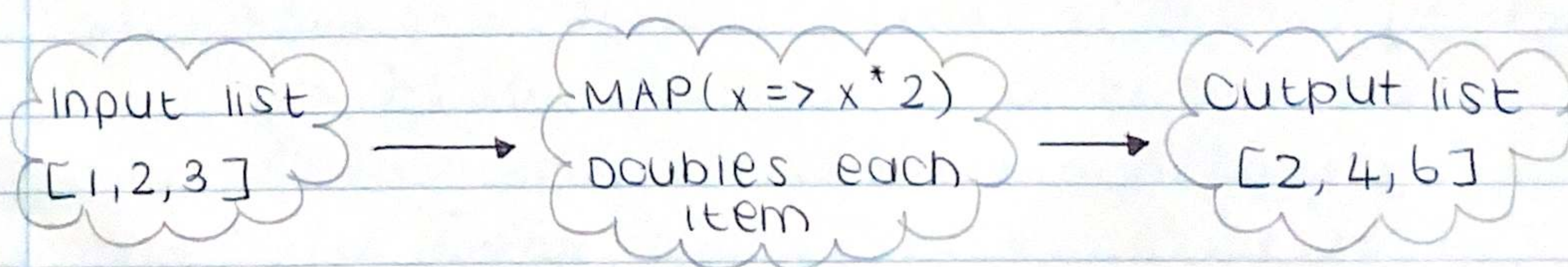
Examples:

1. Poor prompt: "Write about Customer Experience."
Improved: "Write a 200-word essay about how businesses can benefit when customers receive high-quality service, using a professional but easy to understand tone."
2. Poor prompt: "Help me write my CV"
Improved: "Review my CV for a CX role and suggest 3 specific improvements that could make it stronger."

24. A MAP function automatically applies the same action to every item in a list without manually writing a loop. In data processing, it can be used to transform each item in a dataset in the same way.

Example:

```
const numbers = [1, 2, 3];  
const doubled = numbers.map(num => num * 2);  
// doubled = [2, 4, 6]
```



25. cloud computing is using computer resources over the internet instead of physical resources hardware.

► Three main cloud service models:

① IaaS: Provides raw infrastructure as a service (servers and storage)

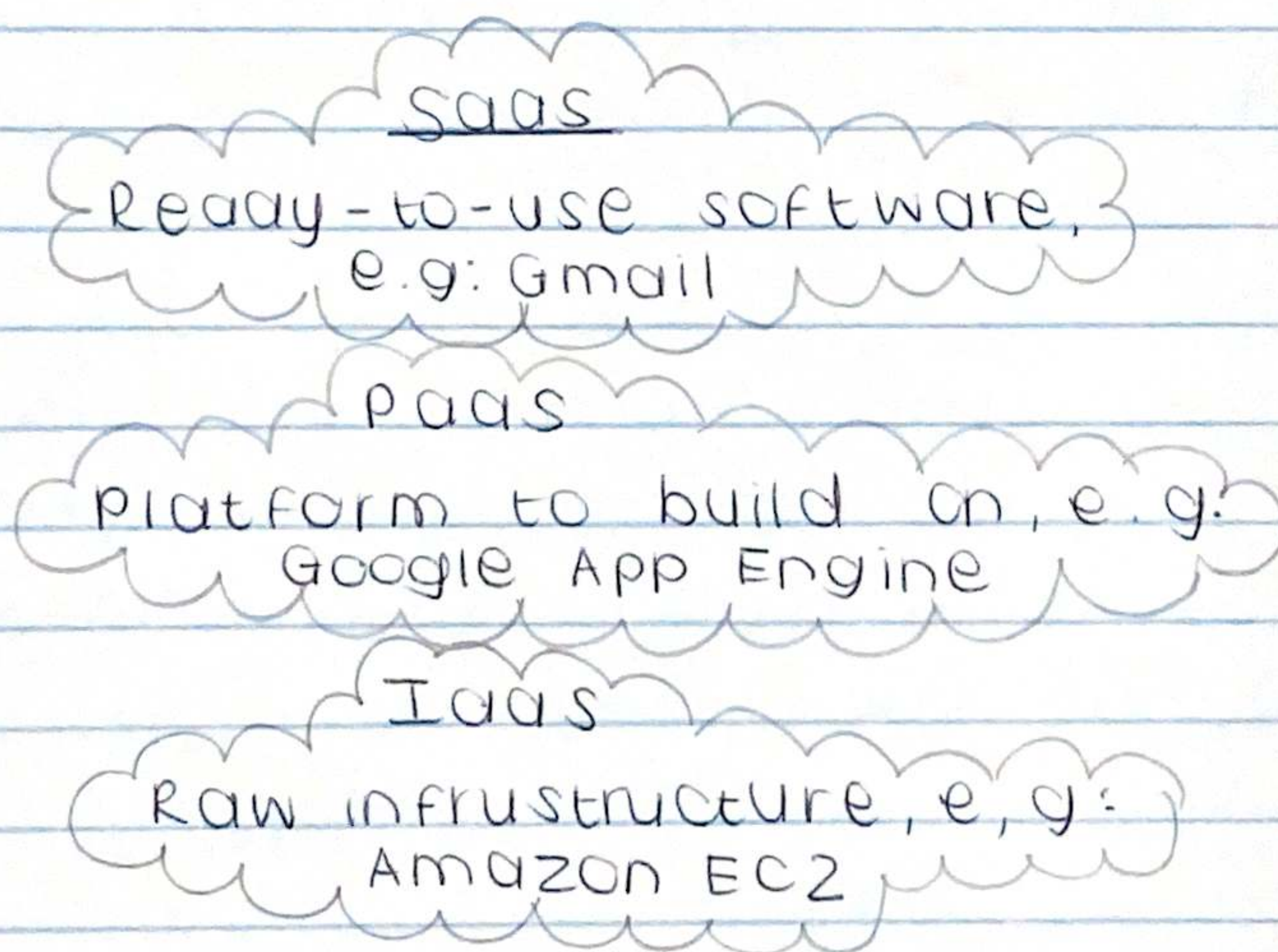
Example: Amazon EC2

② PaaS: Provides a ready-made platform and tools for building and running applications.

Example: Google App Engine

③ SaaS: Provides ready-to-use software over the internet.

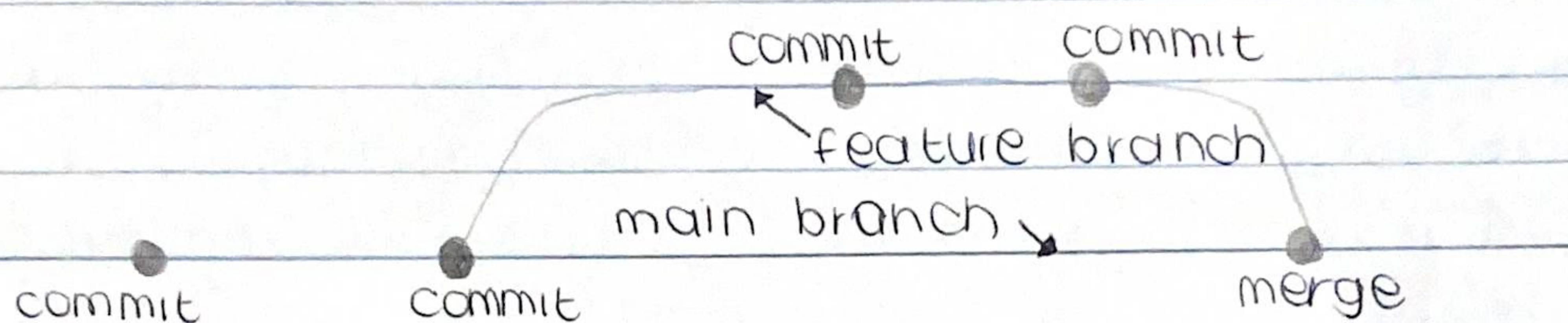
Example: Microsoft 365 and Salesforce



26. A version control system is a tracking system that tracks changes to files over time. Git is a distributed version control system that records modifications made to files in a repository. It is widely used by data analysts and developers because it

allows many people to work on the same project without copying or overwriting each other's work.

A commit is a saved snapshot of changes made at a specific point in time. A branch is a separate line of development where changes can be made without affecting the main branch, and it can later be merged into the main branch.



27. A Jupyter Notebook is a tool commonly used with Python where one can write and run code alongside notes, charts, and results in one place. It is popular among data analysts and data scientists because it allows them to test code and see results immediately in one place.

28. Python is a programming language that's easy to read and write. It is one of the most popular programming languages for data analytics and AI because it is easy to use and has a wide range of existing libraries designed for data work and AI. The four common Python libraries used in data work are:

1. Pandas - For organizing and analyzing table-like data.
2. Numpy - For fast mathematical operations with numbers and arrays.
3. Matplotlib - For creating charts and graphs
4. Scikit-learn - for building simple machine learning models.

► References

1. Postman. (2026) Postman Quick start. Available at: <https://learning.postman.com/docs/getting-started/quick-start> (Accessed: 11 September 2026)
2. Anthropic. 2026 Get started with Claude - API Docs. Available at: <https://docs.anthropic.com/en/docs/get-started> (Accessed: 11 September 2026)
3. AWS. (2026) What is JSON? - JSON Explained. Available at: <https://aws.amazon.com/documentdb/what-is-json> (Accessed: 11 September 2026).
4. Anthropic. (2026) Anthropic's Interactive Prompt Engineering Tutorial. Available at: <https://github.com/anthropics/prompt-eng-interactive-tutorial> (Accessed: 11 September 2026).
5. MDN Web Docs. (2026) Cloud computing - Glossary. Available at: https://developer.mozilla.org/en-US/docs/Glossary/Cloud_computing (Accessed: 11 September 2026).
6. Git. (2026) Git Documentation. Available at: <https://git-scm.com/doc> (Accessed: 12 September 2026)
7. Project Jupyter. (2026) Project Jupyter / Home. Available

at: <https://jupyter.org/> (Accessed : 12 September 2026)

Section D: Data Ethics, Governance & Careers

29. Data Privacy is the protection of ensuring people's personal information is collected, stored and used in the correct manner, where only the right/relevant people can access it. It matters because it builds trust between stakeholders, safeguards people from identity theft and fraud, and is a legal requirement in many countries, as breaking data privacy laws may result in fines.

The General Data Protection Regulation (GDPR) is a data privacy law that was created in 2018 by the European Union, and it gives people control over their own data, where companies are required to get client's permission before collecting any data, provide a reason for collecting said data, allow people to see, rectify, or delete their own data, and lastly report breaches quickly if any are identified concerning their collected data.

GDPR applies to all European Union member countries and can also apply to companies worldwide that handle the personal data of people in the EU, particularly if they offer goods or services to people in the EU or monitor their behaviour.

30. Data Ethics is doing what's fair and morally responsible with data.

Examples of unethical use of data or AI are:

1. Companies selling users personal data without clear consent - we experience this with telecommunications where we receive calls from spam or companies we've never inquired from. This is an issue because we don't agree to have our personal information shared.
2. AI hiring tools rejecting candidates based on age or gender patterns learned from previous company data - This is unfair as it blocks people from getting hired based on their age or gender instead of their actual skills and qualifications.
3. Facial recognition misidentifying people of certain races more often - Innocent people can be wrongfully flagged by police or security systems.

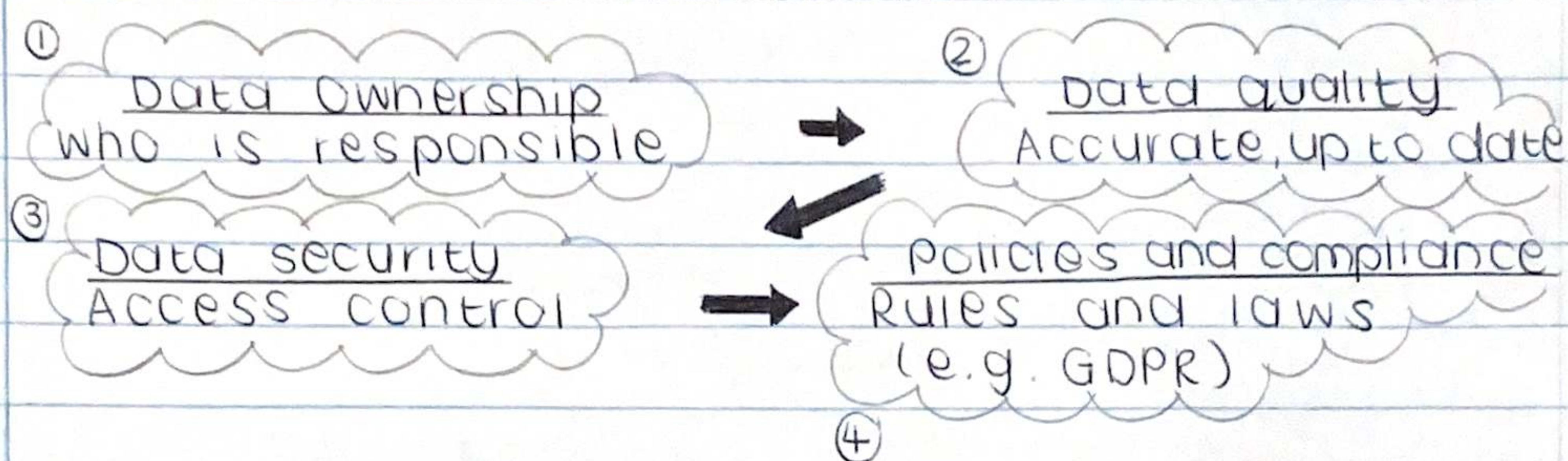
31. Data Governance is a set of rules, roles and processes a company uses to manage its data correctly. It decides who owns and can access the data, and how the quality thereof is maintained.

The key components of a framework are:

- Data ownership - relevant people or departments responsible for each type of data.
- Data quality - ensuring data is accurate, complete and up to date.

- Data security - controls who can access or change data
- Policies and compliance - set rules that follow laws like GDPR

Data Governance Framework



32. AI bias happens when an AI system learns from data that is unbalanced, incomplete or reflects human prejudice, as a result, the system produces unfair decisions.

Real-world example: Amazon built an AI recruiting tool to screen job applications.

It was trained on 10 years of past resumes, most of which came from men because the technology industry has historically been male-dominated. The AI learned to favour male candidates and downgraded applications that included words associated with women. Amazon eventually scrapped the tool.

Potential consequences:

- Qualified people could be unfairly rejected
- The company could lose potentially good and diverse talent
- Existing inequality in society can be reinforced

- The ~~companies~~ company's image can be legally and reputationally damaged.

33. key differences between Data Analyst, Data Scientist, Data Engineer and AI Engineer:

Role	Main Job	key skills	Common Tools
Data Analyst	Examines existing data to find patterns, trends and answers to business questions	Excel, SQL, statistics, reporting	Excel, power BI, Tableau, SQL
Data Scientist	uses data to build models that predict what may happen in the future	statistics, machine learning, programming	python, R, Jupyter, TensorFlow
Data Engineer	creates and maintains systems that collect, store and move data	Database design, coding, cloud systems	SQL, python, Apache, spark, AWS, Azure
AI Engineer	Develops and puts AI systems and models into real-world applications	Machine learning, software engineering, API's	python, PyTorch, Hugging face, cloud AI services

34. Business Intelligence is the process of gathering and transforming raw business data into visual reports and dashboards. It allows managers to make smart, data-based decisions instead of taking simple guesses.

Two popular BI tools:

- Power BI (Microsoft) - connects to company databases and turns data into interactive dashboards and charts

• Tableau - creates very visual, easily comprehensible dashboards. Businesses use it to identify trends quickly, such as best-selling products in a certain region.

BI is essential as it allows managers to quickly identify trends and insights for decision-making instead of digging through spreadsheets to find possible answers.

• References:

1. European Commission / EUR-Lex. General Data Protection (GDPR) - Summary. Available at: <https://eur-lex.europa.eu/eur-lex.europa.eu/legal-content/summary/general-data-protection-regulation-gdpr.html>. (Accessed: 12 September 2026)
2. GDPR.eu. General Data Protection Regulation (GDPR). Available at: <https://gdpr.eu/tag/gdpr/>
3. World Economic Forum. Tech strategy leaders on inclusivity, ethics and responsibility. Available at: <https://www.weforum.org/stories/2024/11/tech-strategy-leaders-inclusivity-ethics-responsibility/>. (Accessed: 12 September 2026)
4. Dastin, J. (2018). Amazon Scraps Secret AI Recruiting Tool showed Bias Against Women. Reuters. Available at: <https://www.reuters.com/article/us-amazon-com-jobs-automationinsight/amazon-scraps-secret-ai-recruiting-tool-that-showed-bias-against-women-idUSKCN1MK08G> (Accessed: 13 September 2026)
5. Microsoft. Power BI. Available at: <https://powerbi.microsoft.com> (Accessed: 12 September 2026)

Section E: Personal Reflection

35. I found AI and Machine Learning the most interesting because I was surprised by how AI can learn patterns from data and use them to make predictions and decisions.
36. I found Data Governance and the different data systems more difficult to understand at first. I used online sources, and simple real-life examples to make the concepts easier to understand.
37. Data and AI can help me make better decisions in my future career by identifying customer trends, analysing business performance, and improve customer experiences. I can also use AI tools to automate repetitive tasks and make more efficiently.

Additional sketch: Three types of Machine Learning

Question 12

① Supervised

Learns from labeled examples



e.g. predicting house prices

② Unsupervised

Find patterns as unlabeled data

e.g. grouping customer segments

③ Reinforcement

Learns by trial and error, rewards

e.g. a game-playing AI improving